Supporting Information for "Attack-Method Choice Changes Robustness Conclusions for a Standard LSTM Rainfall-Runoff Model"

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Text S1. Model Training Configuration

The victim model was trained with the neuralhydrology framework (Kratzert et al., 2022) using the configuration in Table S1. It is a single-layer LSTM with 256 hidden units and a regression head, taking five dynamic Maurer forcing variables and 27 static CAMELS-US catchment attributes. In the Maurer product the daily minimum and maximum temperature inputs are identical, so the five raw channels carry four independent dynamic degrees of freedom. Training used the NSE loss, the Adam optimizer with an initial learning rate of 10^{-3} decayed to 5×10^{-4} at epoch 11 and 10^{-4} at epoch 21, a batch size of 256, gradient-norm clipping at 1.0, and an output dropout of 0.4, for 30 epochs.

The model reproduces the 531-basin temporal benchmark of Kratzert et al. (2019) under the dataset's reverse split: training spans 1 October 1999–30 September 2008 and testing spans 1 October 1989–30 September 1999, over a 270-day input sequence with a single-step (last-day) prediction target. The trained model attains a test-period median NSE of 0.733 across the 531 basins. Adversarial evaluation pins the epoch-30 checkpoint stored under `results/18_lstm_fair_531/lstm_cudalstm_maurer_s100_2026_0616_1513_ep30/` (random seed 100); the per-basin official test metrics used as a reference are read from `test/model_epoch030/test_results.p` within that run directory (Table S3).

Text S2. Basin Coverage and Subsets

All adversarial experiments use the 531 benchmark basin IDs in `src/adversarial/data/531_basins.txt`. The attack-comparison block (Table 3 of the main text), including the best-of-100 random baselines, is run on the full 531 basins; the random-search *scaling* sweep over $K \in \{1, \ldots, 2000\}$ (Text S3) uses a 21-basin subset. The constraint-ablation, targeted-attack, causal-trigger, detectability, and Kling–Gupta reconciliation blocks are run on a 107-basin subset (every fifth basin of that list, taken in list order) to bound computational cost; the minimum-perturbation analysis, which requires an inner bisection over the perturbation radius per basin, uses a 76-basin subset (every seventh basin of that list, taken in list order). Table S2 lists the basin count per block. All 531 basins yield finite NSE and ΔNSE over the test period, so no basins are dropped from any headline statistic.

Table 1 — Model training hyperparameters.

Parameter	Value
Architecture	CudaLSTM (single-layer)
Hidden units	256
Output head	Regression
Training epochs	30
Batch size	256
Sequence length	270 days
Prediction target	Last day (predict_last_n = 1)
Optimizer	Adam (LR $10^{-3} \to 5 \times 10^{-4} \to 10^{-4}$)
Gradient clipping	1.0 (norm)
Output dropout	0.4
Loss function	NSE
Forcing product	Maurer
Dynamic inputs	Precipitation, $T_{\max}$=$T_{\min}$, shortwave radiation, vapor pressure
Static attributes	27 CAMELS-US attributes
Training period	01/10/1999–30/09/2008
Test period	01/10/1989–30/09/1999
Random seed	100

Table 2 — Basin counts per experiment block.

Experiment block	Description	Catchments
Attack comparison	4 methods × 4 ε values	531
Random-search scaling	best-of-K sweep at $\varepsilon = 0.1$	21
Constraint ablation	3 constraints × 3 ε values	107
Targeted attacks	3 targets at $\varepsilon = 0.1$ (NSE + KGE)	107
Causal trigger	4 pre-event windows at $\varepsilon = 0.1$	107
Detectability	KS test at $\varepsilon = 0.1$	107
KGE reconciliation	untargeted attack at $\varepsilon = 0.1$	107
Minimum perturbation	per-basin L_2 safety margin	76

Text S3. Metric Fidelity, Reproducibility, and Random-Search Scaling

Metric fidelity. The attack operates on a single shared per-basin forcing perturbation applied to every overlapping input window, and evaluates the NSE on the concatenated last-day predictions. This metric is constructed to match the official benchmark per-basin NSE that the neuralhydrology tester reports. Across all 531 basins and four budgets, the attack's clean NSE agrees with the official tester's per-basin NSE to within a maximum absolute difference of 2.65×10^{-7}, confirming that the adversarial degradations are measured on the same metric used to report clean skill.

Random-search scaling. The main text states that a best-of-K random search cannot close the gradient-versus-random gap even when extrapolated to very large sample budgets. We verify this on the 21-basin scaling subset by sweeping the number of random-sign draws $K \in \{1, 3, 10, 30, 100, 300, 1000, 2000\}$ and recording the median best-of-K degradation. The median random degradation grows approximately as $a\sqrt{\ln K} + b$ (least-squares fit over $K \geq 10$, giving $a = 0.0117$, $b = -0.0031$, coefficient of determination above 0.99). Extrapolating this fit to $K = 10^9$ gives a median random degradation of about 0.050, against a median scheduled-step-PGD degradation of 0.342 on the same basins—so even a billion-sample random search would remain roughly seven times weaker than the gradient attack. The gradient-versus-random gap is therefore not an artifact of an under-resourced random baseline.

Reproducibility. Table S3 lists the artifacts needed to reproduce the analysis. The attack driver and per-experiment runners are tracked in the repository; the per-basin result records are written as JSON under the output directory and aggregated by a single script that prints every number reported in the main text.

Table 3 — Reproducibility artifacts.

Artifact	Path or value
Victim run	`results/18_lstm_fair_531/lstm_cudalstm_maurer_s100_2026_0616_1513_ep30/`
Pinned checkpoint	`test/model_epoch030/` (epoch 30)
Reference test metrics	`test/model_epoch030/test_results.p`
Basin list	`src/adversarial/data/531_basins.txt`
Attack driver	`src/adversarial/scripts/coherent_attack.py`
Experiment runners	`src/adversarial/scripts/run_exp{1..6}.py`, `run_exp_detect.py`
Output records	`results/05_adversarial_robustness/id18_s100/exp*.json`, `*_summary.csv`
Aggregator	`src/adversarial/scripts/aggregate_id18.py`
Hydrological attribution	`analyze_hydro_vulnerability.py`, `analyze_l1_attribution.py`,
	`analyze_l2_signatures.py`, `analyze_drought_mechanism.py` (in `src/adversarial/scripts/`)
Metric-fidelity check	$\max \lvert \mathrm{NSE}_{\mathrm{clean}} - \mathrm{NSE}_{\mathrm{official}} \rvert = 2.65 \times 10^{-7}$ (531 basins)
Finite NSE metrics	531 of 531 basins

Text S4. Hydrological Attribution and the Wet–Dry Gradient

This section gives the full statistics behind the catchment-attribute analysis (Section 3.2 of the main text) and its identifiability checks. All partial Spearman correlations rank-transform each variable, regress the rank-transformed controls out of both the response and the predictor by ordinary least squares, and report the Pearson correlation of the residuals; p-values use $df = N - 2$. Regime labels follow the main text: a basin is *snow* if its snow fraction exceeds 0.2, and among the remaining basins *dry* if its aridity (potential evapotranspiration over precipitation) exceeds 1.5 and *humid* otherwise. Every value below is printed by `analyze_drought_mechanism.py`.

The damage gradient with catchment character (Table 4) is real but not separable into a distinct drought mechanism. The climate and flow-magnitude descriptors are mutually correlated at $|\rho| = 0.60$–0.94, and additionally controlling for mean specific discharge flips the aridity partial from $+0.18$ to -0.13 ($p = 3.4 \times 10^{-3}$) and collapses the gradient-to-random damage ratio's aridity partial from $+0.23$ to -0.02 ($p = 0.70$). The mean-precipitation partial itself weakens from -0.37 under clean-skill control alone to -0.26 once snow fraction is also held fixed, showing that part of the precipitation gradient is snow-mediated. Consistent with the absence of a distinct dry-end mechanism, the attack does not re-aim its perturbation toward the dry end: within the non-snow population the precipitation share of the damage tends to rise only weakly with aridity (+0.36, $N = 71$), and neither the flood-peak signature (-0.20, $p = 0.09$) nor the low-flow-share signature (-0.10, $p = 0.40$) is significant there. The two significant full-subset dry-end partials (peak magnitude -0.28; low-flow share -0.23) are driven by snow leakage—within the snow basins alone the peak-magnitude–aridity Spearman correlation is -0.55 ($p = 5 \times 10^{-4}$)—which is why the main-text nulls are scoped to the non-snow population. Only seven dry non-snow basins fall in the 107-basin subset, so these nulls exclude only large effects.

Table 4 — Wet–dry damage gradient and identifiability checks. Damage is $-\Delta$NSE under scheduled-step PGD at $\varepsilon = 0.1$; $D_{\mathrm{APGD}}/D_{\mathrm{rand}}$ is the per-basin ratio of gradient to best-of-100 random-sign damage. Partial Spearman correlations control for snow fraction and clean NSE unless noted.

Quantity	Predictor / control	Partial ρ (p)
Damage	aridity	$+0.18$ (2.9×10^{-5})
Damage	mean precipitation	-0.26 (1.5×10^{-9})
Damage	runoff ratio	-0.25 (4.5×10^{-9})
Damage	mean specific discharge	-0.28 (7.7×10^{-11})
Damage	baseflow index	$+0.13$ (2.4×10^{-3})
Damage	mean precip. (clean skill only)	-0.37 (6.3×10^{-19})
Damage	aridity (+ mean sp. discharge)	-0.13 (3.4×10^{-3})
$D_{\mathrm{APGD}}/D_{\mathrm{rand}}$	aridity (+ mean sp. discharge)	-0.02 (0.70)
Precip. damage share	aridity (non-snow, $N = 71$)	$+0.36$ (2.0×10^{-3})
Peak-magnitude change	aridity (non-snow, $N = 71$)	-0.20 (0.09)
Low-flow damage share	aridity (non-snow, $N = 71$)	-0.10 (0.40)

References

Kratzert, F., Gauch, M., Nearing, G., & Klotz, D. (2022). NeuralHydrology—A Python library for deep learning research in hydrology. *Journal of Open Source Software*, *7*(71), 4050.

Kratzert, F., Klotz, D., Shalev, G., Klambauer, G., Hochreiter, S., & Nearing, G. (2019). Towards learning universal, regional, and local hydrological behaviors via machine learning applied to large-sample datasets. *Hydrology and Earth System Sciences*, *23*(12), 5089–5110.